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Marie Bashir Institute
Tackling infections, locally and globally



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Exercise 4: Deciphering transmission pathways

Summer School in Pathogen
Genomics and Global Health

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Introduction

In this exercise, we will analyse an outbreak of Methicillin-resistant *Staphylococcus aureus* (MRSA) in a special care baby unit (Harris et al., 2013. *Lancet Infectious Diseases* 13:130-136). We will run a pan genome bioinformatics pipeline on an online Genomics Virtual Lab, Galaxy <https://galaxy-mel.genome.edu.au/galaxy/> and explore the results of the pipeline. We will inspect the quality of whole genome sequencing reads for one MRSA isolate; view the annotated draft genome (assembled using de novo assembly); identify the core and accessory genome; and, draw the phylogeny of a subset of isolates from the outbreak.

Case study

Whole-genome sequencing for analysis of an outbreak of meticillin-resistant *Staphylococcus aureus*: a descriptive study

Simon R Harris*, Edward J P Cartwright*, M Estée Török, Matthew T G Holden, Nicholas M Brown, Amanda L Ogilvy-Stuart, Matthew J Ellington, Michael A Quail, Stephen D Bentley, Julian Parkhill†, Sharon J Peacock†

Summary

Background The emergence of meticillin-resistant *Staphylococcus aureus* (MRSA) that can persist in the community and replace existing hospital-adapted lineages of MRSA means that it is necessary to understand transmission dynamics in terms of hospitals and the community as one entity. We assessed the use of whole-genome sequencing to enhance detection of MRSA transmission between these settings.

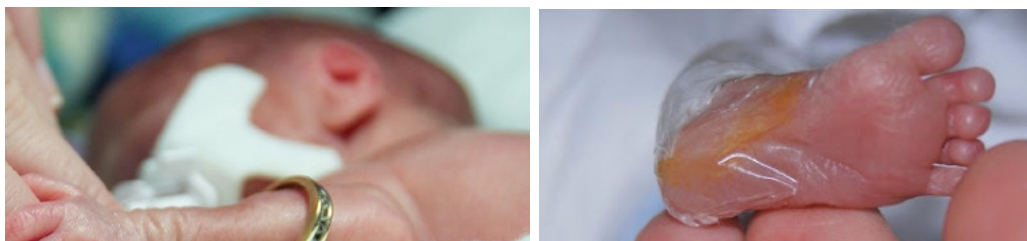
Methods We studied a putative MRSA outbreak on a special care baby unit (SCBU) at a National Health Service Foundation Trust in Cambridge, UK. We used whole-genome sequencing to validate and expand findings from an infection-control team who assessed the outbreak through conventional analysis of epidemiological data and antibiogram profiles. We sequenced isolates from all colonised patients in the SCBU, and sequenced MRSA isolates from patients in the hospital or community with the same antibiotic susceptibility profile as the outbreak strain.

Findings The hospital infection-control team identified 12 infants colonised with MRSA in a 6 month period in 2011, who were suspected of being linked, but a persistent outbreak could not be confirmed with conventional methods. With whole-genome sequencing, we identified 26 related cases of MRSA carriage, and showed transmission occurred within the SCBU, between mothers on a postnatal ward, and in the community. The outbreak MRSA type was a new sequence type (ST) 2371, which is closely related to ST22, but contains genes encoding Panton-Valentine leucocidin. Whole-genome sequencing data were used to propose and confirm that MRSA carriage by a staff member had allowed the outbreak to persist during periods without known infection on the SCBU and after a deep clean.

Interpretation Whole-genome sequencing holds great promise for rapid, accurate, and comprehensive identification of bacterial transmission pathways in hospital and community settings, with concomitant reductions in infections, morbidity, and costs.

Lancet Infectious Diseases, 2013, 13(2):130-136

[https://dx.doi.org/10.1016/S1473-3099\(12\)70268-2](https://dx.doi.org/10.1016/S1473-3099(12)70268-2)



Background

A potential outbreak of MRSA in the Special Care Babies Unit (SCBU) was indicated by three babies having similar antibiotic susceptibility profiles and who were admitted within days of each other.

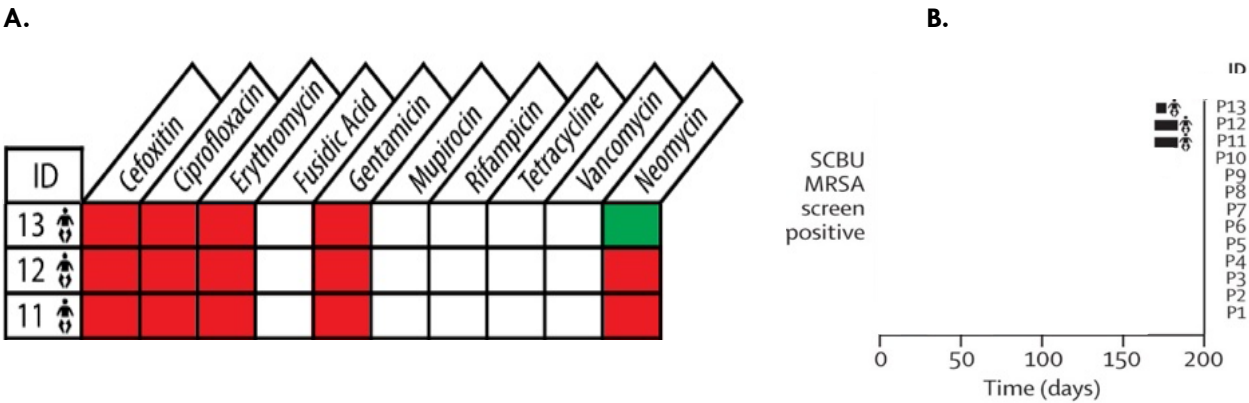


Figure 1. (A) Antibigram and (B) epidemiology of three possible outbreak cases in the SCBU.

Infection control responded to the outbreak by performing a deep clean, implemented weekly MRSA screening, and reinforced infection control. 13 (+3) babies with MRSA were identified within 6 months prior to the 3 outbreak cases. 1 case was identified after the deep clean of the SCBU. Infection control investigated 17 cases: 12 cases were classified as part of the outbreak and 5 cases were classified as unrelated to the outbreak.

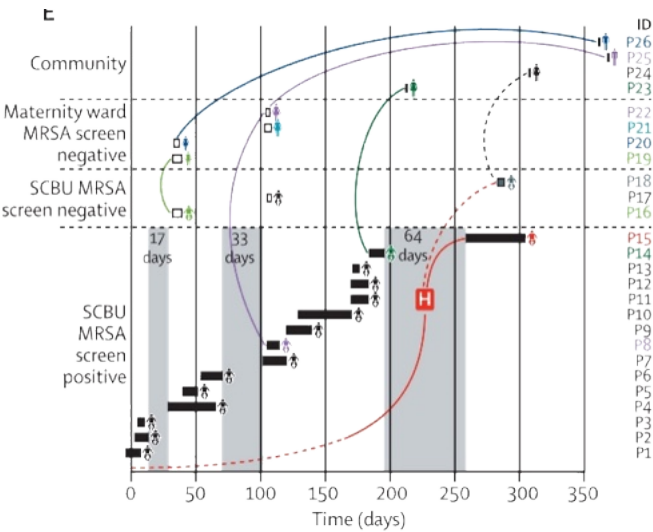


Figure 2. Epidemiology of patients in the SCBU and other patients with linked MRSA infection detected in the community.

Whole Genome Sequencing

Did the outbreak extend over 6 months?

How accurate was the infection control investigation?

Reference mapping

(approach used by Harris et al.)

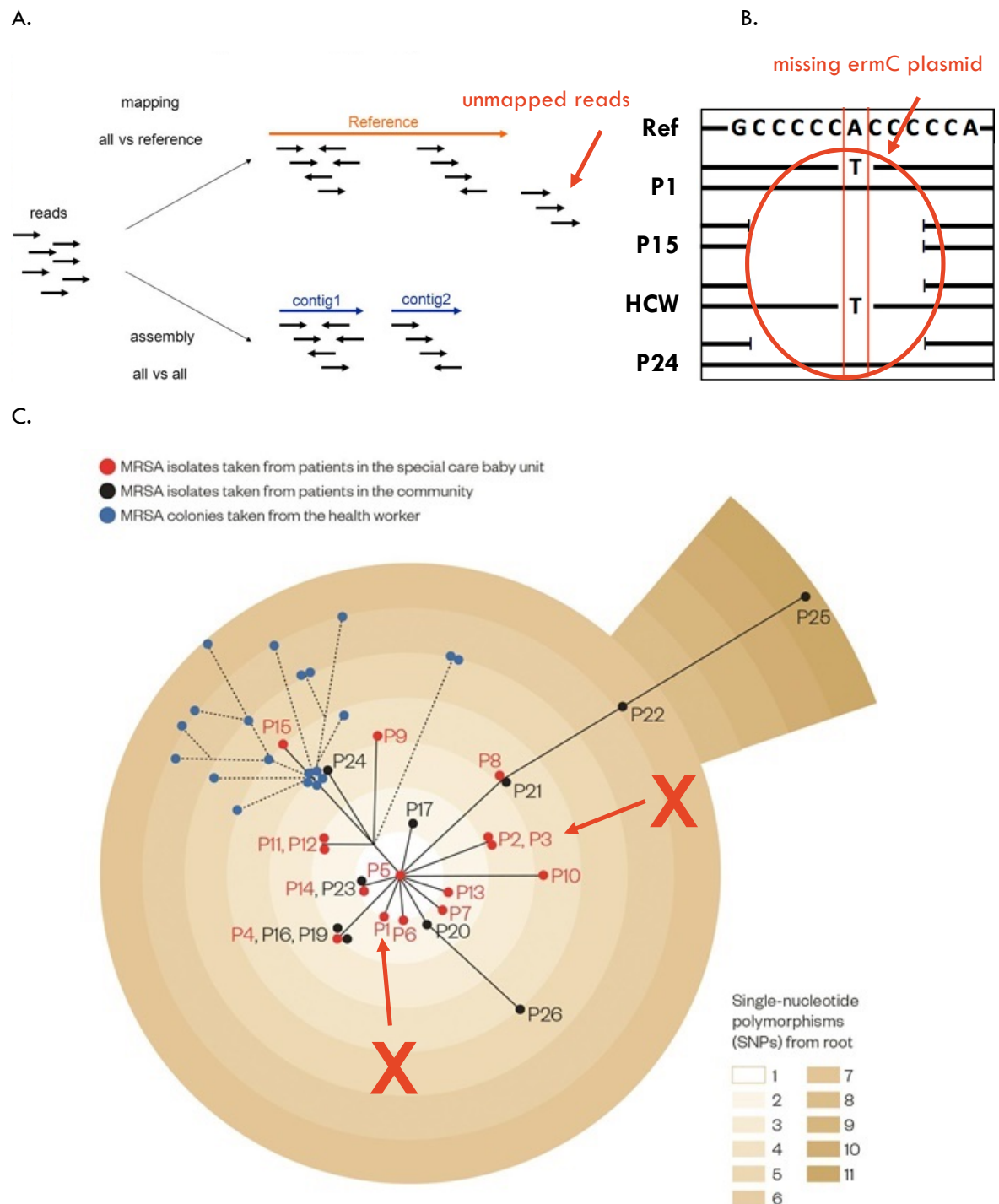
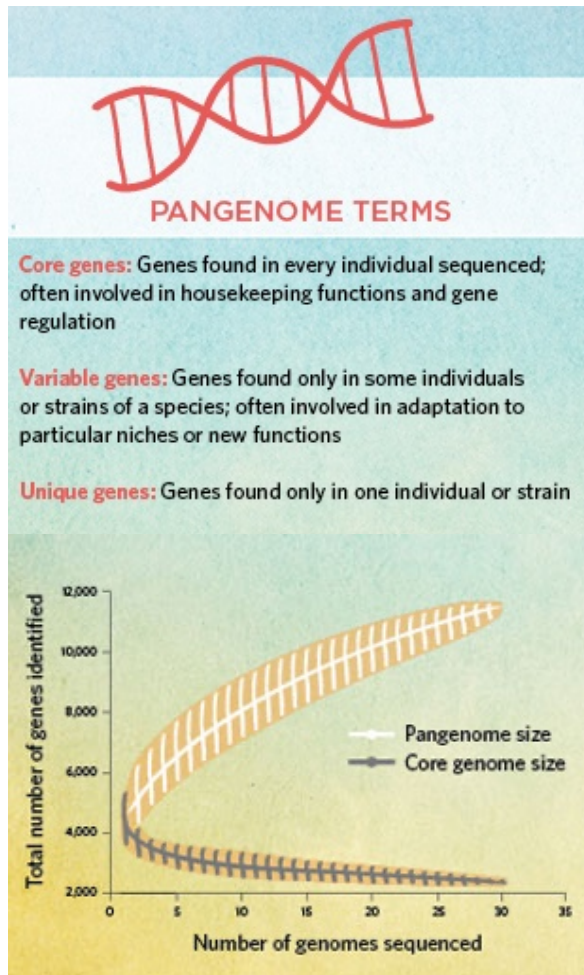


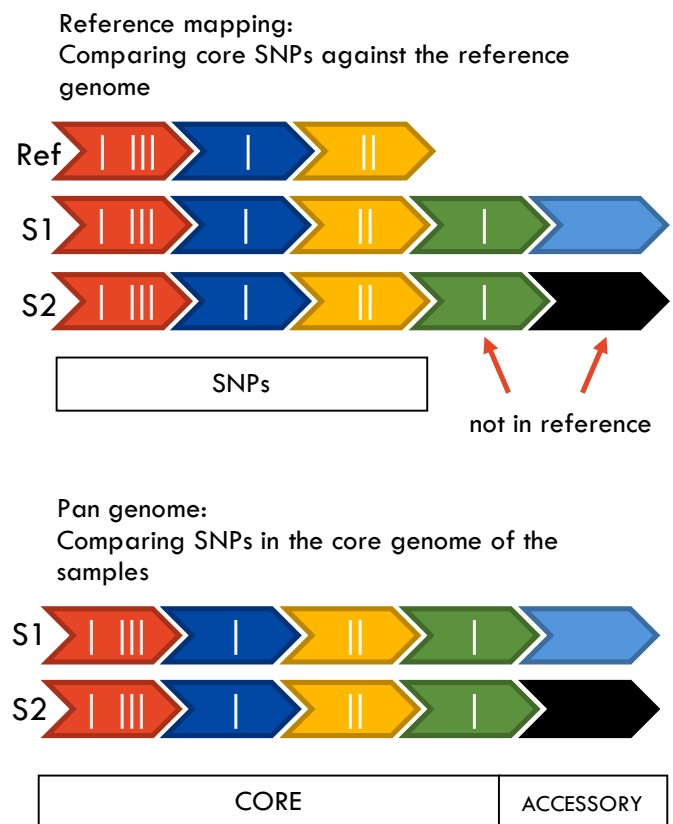
Figure 3. (A) Creating a draft genome by reference mapping vs de novo assembly. **(B)** A deletion found in samples when compared to the reference. **(C)** Phylogenetic tree based on core single nucleotide polymorphisms (SNPs), including P1 and P3 who infection control had initially excluded from the outbreak based on their antibiogram.

The pan genome

A.



B.



Offord, 2016. Infographic: Partitioning the Genome.

<http://www.the-scientist.com/?articles.view/articleNo/47561/title/Infographic--Partitioning-the-Genome/>

Figure 4. (A) Pan genome definition and a visualisation of the size of a species's core genome decreasing and the size of the pangenome increasing as more sequenced genomes are added. (B) Comparison of genomes by reference mapping and by pan genome.

Launch the pipeline

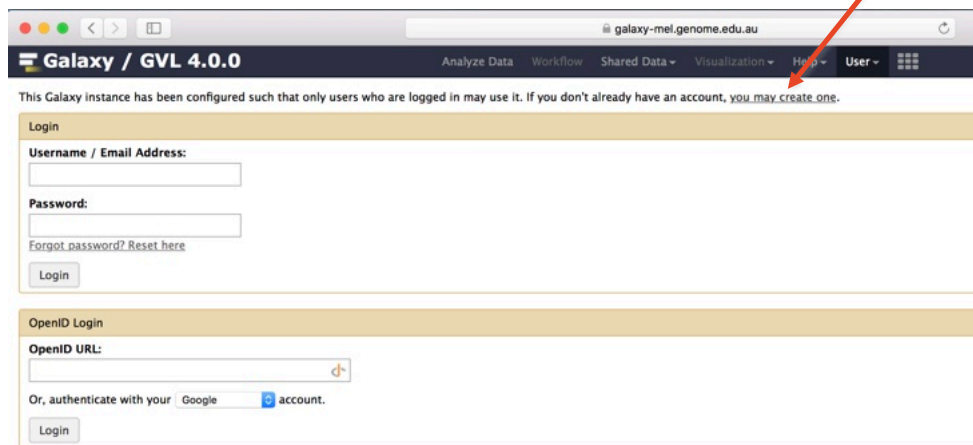
Galaxy is an open source, web-based *research cloud* for life sciences and biomedical research.

The Galaxy service we will use today is maintained by the Life Sciences Computation Centre at the VLSCI who provide this Galaxy instance to:

- provide bioinformatics infrastructure to the life sciences community
- give researchers tools to do their own analysis
- help improve general bioinformatics skills
- provide a platform for tutorials and other training

Create your account

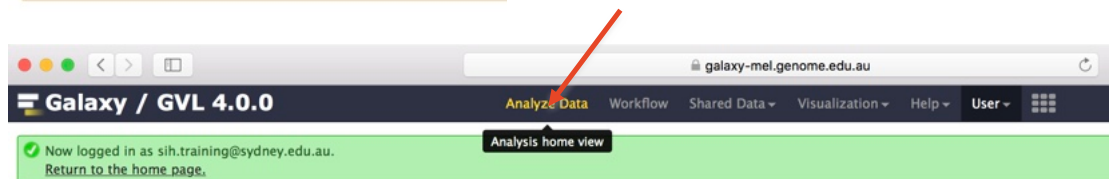
1. Log into <https://galaxy-mel.genome.edu.au>



A screenshot of a web browser showing the Galaxy login page. The browser's address bar displays 'galaxy-mel.genome.edu.au'. The page header includes 'Galaxy / GVL 4.0.0' and navigation links: 'Analyze Data', 'Workflow', 'Shared Data', 'Visualization', 'Help', and 'User'. A message states: 'This Galaxy instance has been configured such that only users who are logged in may use it. If you don't already have an account, you may create one.' Below this is a 'Login' section with fields for 'Username / Email Address' and 'Password', a 'Forgot password? Reset here' link, and a 'Login' button. An 'OpenID Login' section follows, with an 'OpenID URL' field and a 'Login' button. A note says 'Or, authenticate with your Google account.' with a Google icon.



A screenshot of the Galaxy account creation page. The header shows 'Galaxy / GVL 4.0.0'. The 'Create account' section contains fields for 'Email address' (filled with 'rosemarie.sadsad@sydney.edu.au'), 'Password' (masked with dots and a strength indicator), 'Confirm password' (masked with dots), and 'Public name' (filled with 'rosemarie.sadsad'). A note explains: 'Your public name is an identifier that will be used to generate address: underscores, and dashes (',', '_', '-').'. A 'Submit' button is at the bottom.



A screenshot of the Galaxy home page after successful login. The browser address bar shows 'galaxy-mel.genome.edu.au'. The header includes 'Galaxy / GVL 4.0.0' and navigation links: 'Analyze Data', 'Workflow', 'Shared Data', 'Visualization', 'Help', and 'User'. A green banner at the bottom states: 'Now logged in as sih.training@sydney.edu.au. Return to the home page.' A button labeled 'Analysis home view' is visible below the banner.

Import your data

1. Open the public histories:

https://galaxy-mel.genome.edu.au/galaxy/history/list_published

2. Click on mrsa-summer-school-2018

Galaxy / GVL 4.0.0 Analyze Data Workflow Shared Data Visualization

Published Histories

search name, annotation, owner, and tags Q

Advanced Search

Name	Annotation	Owner
mrsa-summer-school-2018		sih.training

You can also access the mrsa-summer-school-2018 dataset directly:

<http://galaxy-mel.genome.edu.au/galaxy/u/sih.training/h/mrsa-summer-school-2018>

3. Import the dataset into your "history"

Galaxy / GVL 4.0.0 Analyze Data Workflow Shared Data Visualization Help User

Published Histories | sih.training | mrsa-summer-school-2018

mrsa-summer-school-2018
2.17 GB

search datasets

Dataset Annotation

10: P24_2.fastq	
9: P24_1.fastq	

Import History

Make a copy of this history and switch to it

sih.training

Relate All publ Publish Rating Commu (0 ratings)

Importing history "mrsa-summer-school-2018"

Enter a title for the new history:

imported: mrsa-summer-school-2018

Cancel Import

Galaxy / GVL 4.0.0 Analyze Data Workflow Shared Data Visualization Help User

Tools

search tools

BASIC TOOLS

- Get Data
- Send Data
- Lift-Over
- Text Manipulation
- Filter and Sort
- Join, Subtract and Group
- Convert Formats
- Operate on Genomic Intervals
- Extract Features

STATISTICS AND VISUALISATION

- Fetch Sequences
- Fetch Alignments
- Statistics
- Graph/Display Data

NGS COMMON TOOLSETS

- FASTA manipulation
- NGS: Picard
- NGS: SAM Tools

Welcome to Galaxy Melbourne

This Galaxy service is maintained by the Life Sciences Computation Centre at the VLSCI

We provide this Galaxy instance to:

- provide bioinformatics infrastructure to the Melbourne life sciences community
- give researchers tools to do their own analysis
- help improve general bioinformatics skills
- provide a platform for tutorials and other training

IMPORTANT NOTE:

Disk Quotas
Registered users get 100GB of disk space.
If you require more space, please contact the service administrators by email

History

search datasets

Imported: mrsa-summer-school-2018
10 shown

2.17 GB

10: P24_2.fastq		
9: P24_1.fastq		
8: P15_2.fastq		
7: P15_1.fastq		
6: P5_2.fastq		
5: P5_1.fastq		
4: P1_2.fastq		
3: P1_1.fastq		
2: HCW_2.fastq		
1: HCW_1.fastq		

Import the bioinformatics pipelines

1. Go to the publicly available workflows

https://galaxy-mel.genome.edu.au/galaxy/workflow/list_published

2. Import both the **annotation-summer-school-2018** workflow and the **pangenome-summer-school-2018** workflow.

Galaxy / GVL 4.0.0

Analyze Data Workflow Shared Data Visualization Help User

Published Workflows

search name, annotation, owner, and tags

Advanced Search

Name	Annotation	Owner
pangenome-summer-school-2018	Pathogen Genomics Summer School 2018	sih.training
annotation-summer-school-2018	Pathogen Genomics Summer School 2018	sih.training

Galaxy / GVL 4.0.0

Analyze Data Workflow

Workflow "pangenome-summer-school-2018" has been imported.
You can [start using this workflow](#) or [return to the previous page](#).

3. Click on Workflow in the menu to go to your workflows

Galaxy / GVL 4.0.0

Analyze Data Workflow Shared Data

Chain tools into workflows

Workflow "annotation-summer-school-2018" has been imported.
You can [start using this workflow](#) or [return to the previous page](#).

Galaxy / GVL 4.0.0

Analyze Data Workflow

Your workflows

Name
imported: annotation-summer-school-2018
imported: pangenome-summer-school-2018

Run the pipelines

1. Run the annotation-summer-school-2018 workflow

Galaxy / GVL 4.0.0

Your workflows

Name

imported: annotation-summer-school-2018 ▼

imported: pangenome-summer-school-2018 ▼

Edit

Run

2. Select the data to be analysed.

Click on the multiple datasets button



for '1: Input dataset' and '2: Input dataset'

For '1: Input dataset', use the control button to select the 5 x forward read files ending in _1.fastq

For '2: Input dataset', use the control button to select the 5 x reverse read files ending in _2.fastq

3. Click Run workflow

Workflow: annotation-summer-school-2018

Run workflow

History Options

Send results to a new history

Yes No

1: Input dataset

4: P1_2.fastq

3: P1_1.fastq

Multiple datasets: CW_2.fastq

1: HCW_1.fastq

This is a batch mode input field. Separate jobs will be triggered for each dataset selection.

Batch options:

2: Input dataset

4: P1_2.fastq

3: P1_1.fastq

2: HCW_2.fastq

1: HCW_1.fastq

This is a batch mode input field. Separate jobs will be triggered for each dataset selection.

Batch options:

3: FastQC Read Quality reports (Galaxy Version 0.67)

Short read data from your current history

History

search datasets

imported: mrsa-summer-school-2018

10 shown

2.17 GB

10: P24_2.fastq

9: P24_1.fastq

8: P15_2.fastq

7: P15_1.fastq

6: P5_2.fastq

5: P5_1.fastq

4: P1_2.fastq

3: P1_1.fastq

2: HCW_2.fastq

1: HCW_1.fastq

Successfully invoked workflow annotation-summer-school-2018 5 times.

You can check the status of queued jobs and view the resulting data by refreshing the History pane. When the job has been run the status will change from 'running' to 'finished' if completed successfully or 'error' if problems were encountered.

History

search datasets

imported: mrsa-summer-school-2018

127 shown

2.17 GB

130: Prokka on data

117: log

129: Prokka on data

117: txt

4. Once the annotation pipeline has completed, run the pangenome-summer-school-2018 workflow

Your workflows

Name

pangenome-summer-school-2018 ▾

annotation-summer-school-2018 ▾

Workflows shared by others

No workflows have been shared by others

Other options

Configure your workflow menu

- Edit
- Run
- Share or Download
- Copy
- Rename
- View
- Delete

2. Select the data to be analysed:

Hold control and select the 5 gene feature files.

Click 'Run workflow'

Workflow: pangenome-summer-school-2018

✓ Run workflow

History Options

Send results to a new history

☐ Yes ☐ No

1: Roary the pangenome pipeline – Quickly generate a core gene alignment from gff3 files (Galaxy Version 3.10.2)

Individual gff files or a dataset collection

Individual

select gff inputs to Roary

96: Prokka on data 93: gff
72: Prokka on data 69: gff
48: Prokka on data 45: gff
24: Prokka on data 21: gff

Select the files you wish to send to Roary, must be in gff3 format with the sequence data at the end of the file.

☒ **minimum percentage identity for blastp**

95

☒ **percentage of isolates a gene must be in to be core**

99.0

☒ **Additional outputs**

Accessory binary genes in newick format Accessory binary genes in fasta format Accessory gene table in tabular format Gene Presence Absence file in Rtab format Number of Conserved Genes in Rtab format Number of Genes in Pan Genome in Rtab format Number of New Genes in Rtab format Number of Unique Genes in Rtab format

Advanced options



Successfully invoked workflow `pangenome-summer-school-2018`.

You can check the status of queued jobs and view the resulting data by refreshing the History pane. When the job has been run the status will change from 'running' to 'finished' if completed successfully or 'error' if problems were encountered.

History

search datasets

imported: mrsa-summer-school-2018

52 shown, 90 hidden

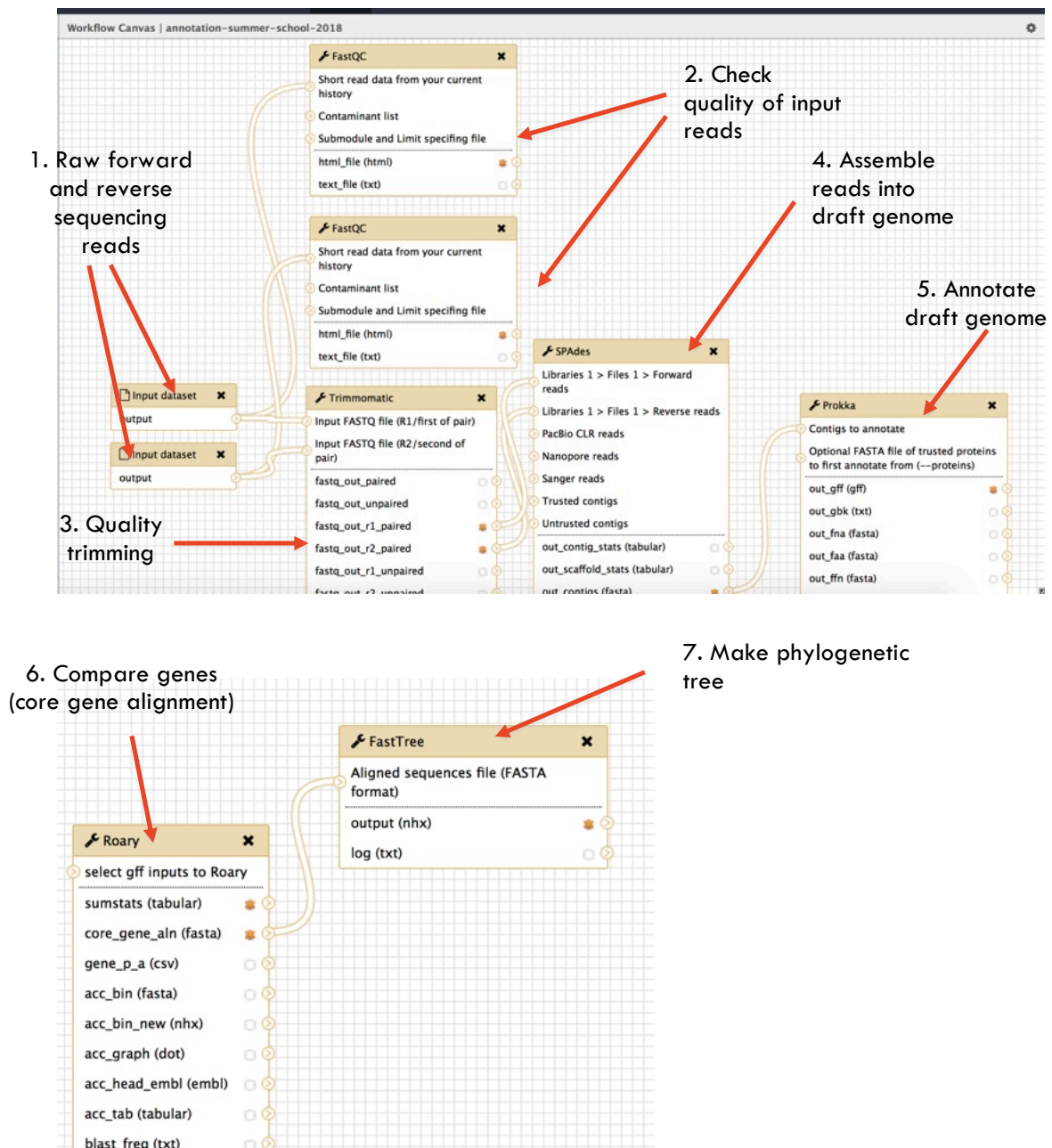
4.36 GB

142: FastTree on data_132-tree.nhx

141: Roary on data_120, data_96, and others

Number of Unique Genes

The pangenome pipeline



Explore the pipeline results

We have become familiar with the case study and have learned about de novo assembly, the pan genome, and a bioinformatics pan genome pipeline. In this time, the pan genome pipeline that we started at the beginning of the class has been working hard and producing data. Let's explore this data.

Read quality

We will check the quality of the input sequencing reads.

1. Look for the FastQC on data 1: Webpage, click on the eye  to view the results.

HCW_1.fastq FastQC Report
FastQC Report
Mon 12 Feb 2018
HCW_1.fastq

Summary

- ✓ Basic Statistics
- ✓ Per base sequence quality
- ✓ Per tile sequence quality
- ✓ Per sequence quality scores
- ✗ Per base sequence content

History

search datasets

Imported: mrsa-summer-school-2018
52 shown, 91 hidden

4.36 GB

W 2.fastq (R2 paired)

15: Trimmomatic on HC W 1.fastq (R1 paired)

13: FastQC on data 2: Webpage

11: FastQC on data 1: Webpage

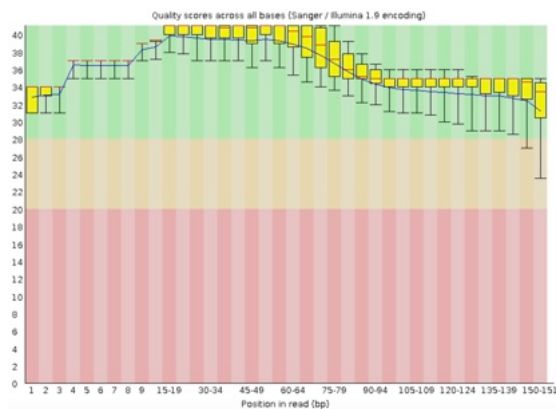
10: P24 2.fastq

View data

Basic Statistics

Measure	Value
Filename	HCW_1.fastq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	948874
Sequences flagged as poor quality	0
Sequence length	151
%GC	33

Per base sequence quality

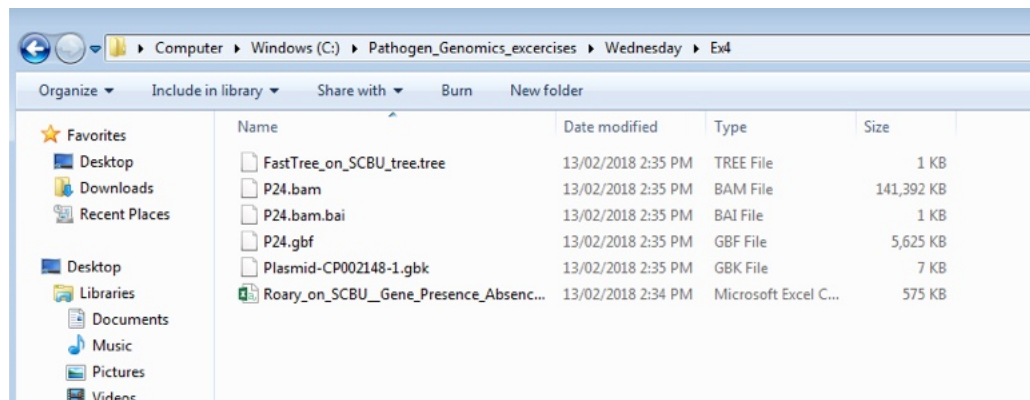


Phylogeny and genome comparison

We will view the phylogenetic tree drawn by the pan genome pipeline and compare the genomes of five isolates from the outbreak of MRSA in the SBCU.

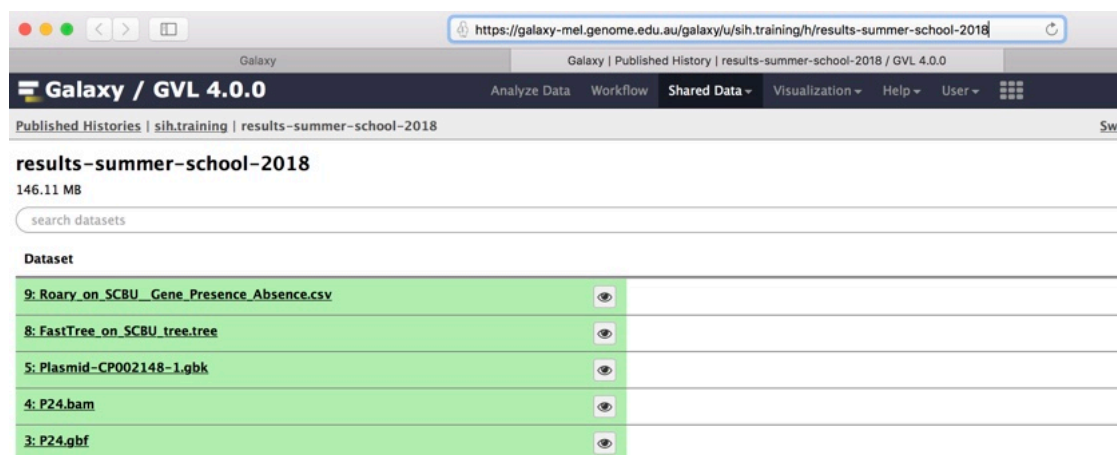
The data for the next section can be found in:

C:\Pathogen_Genomics_exercises\Wednesday\Exercise4

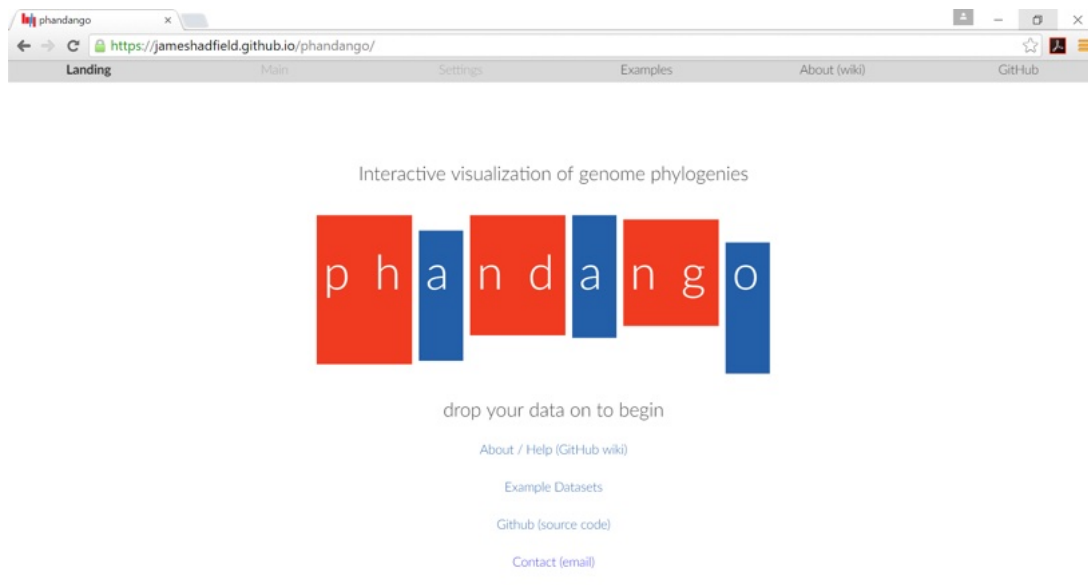


The data is also publicly available here:

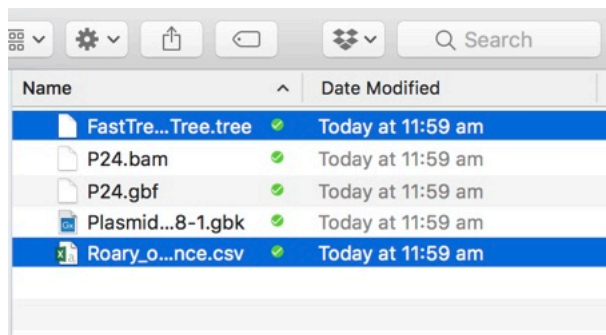
<https://galaxy-mel.genome.edu.au/galaxy/u/sih.training/h/results-summer-school-2018>



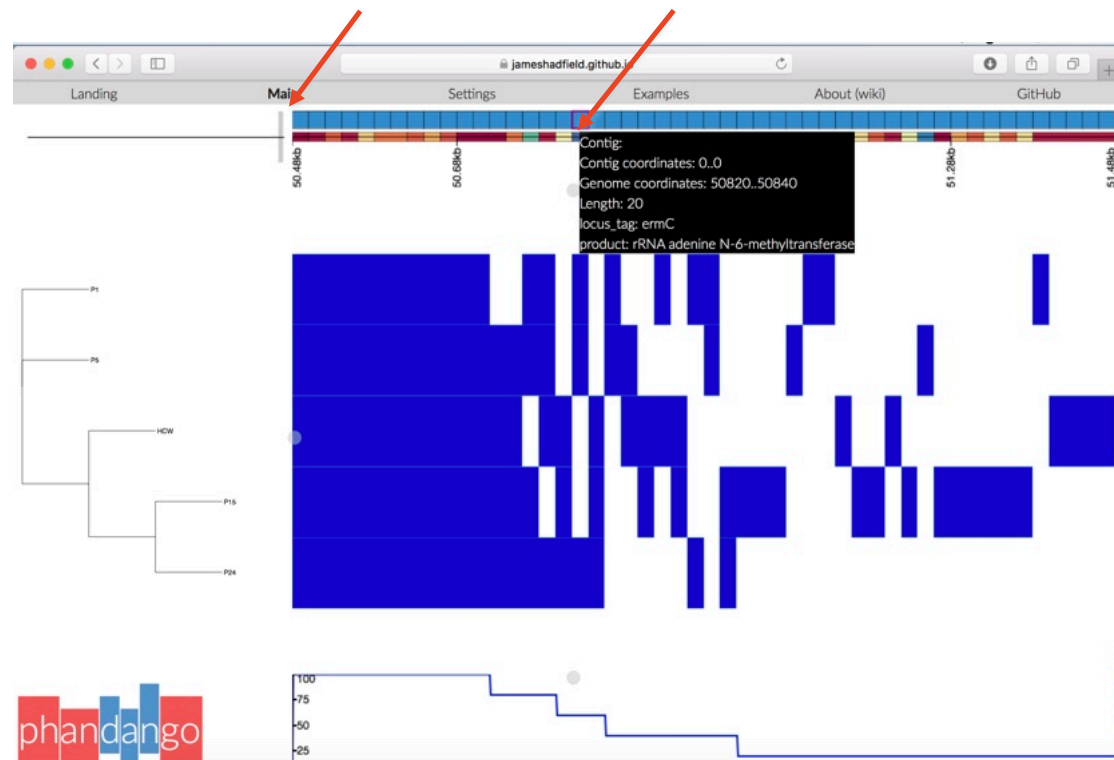
1. Open the Chrome  web browser (**NOT** Internet Explorer) and go to <https://jameshadfield.github.io/phandango/>



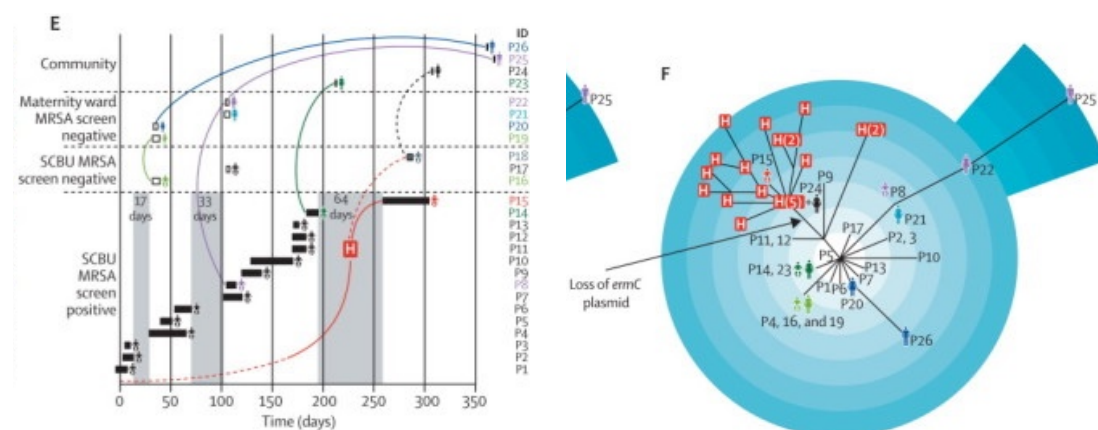
2. In the Exercise4 folder, select the **FastTree_on_SCBU_tree.tree** phylogenetic tree and the **Roary_on_SCBU_Gene_Presence_Absence.csv** gene presence and absence file and drag these into the Phandango webpage.



3. Zoom into the heat map, then move to the accessory genes (end of the page). Hover over the top blue squares looking for the *ermC* gene (rRNA adenine N-6-methyltransferase, confers Erythromycin resistance).



See how the phylogeny compares with that reported in the paper.



Notice that 3 samples, P1, P5, and P24 have the *ermC* gene. However, the paper reports that P24 was susceptible to Erythromycin and was missing the plasmid carrying *ermC*.

Table S1. Antimicrobial susceptibility pattern of MRSA isolates.

ID	Cefoxitin	Ciprofloxacin	Erythromycin	Fusidic Acid	Gentamicin	Mupirocin	Rifampicin	Tetracycline	Vancomycin	Neomycin
	Red	Red	Blue	Red	White	White	White	White	White	Red
24 	Red	Red	White	Red	White	White	White	White	White	Red
15 	Red	Red	White	Red	White	White	White	White	White	Red
5 	Red	Red	Red	Red	White	White	White	White	White	Red
1 	Red	Red	Red	Green	White	White	White	White	Green	White

Infant Healthcare worker 

Red means resistant and white means susceptible to a given antibiotic. Green, initial result proved incorrect on repeat testing and changed designation from susceptible to resistant. A healthcare worker had twenty MRSA colonies taken from the primary plate; the blue box for erythromycin signifies that some colonies were susceptible to this antibiotic (n=18) and some were resistant (n=2).

Let’s explore this.

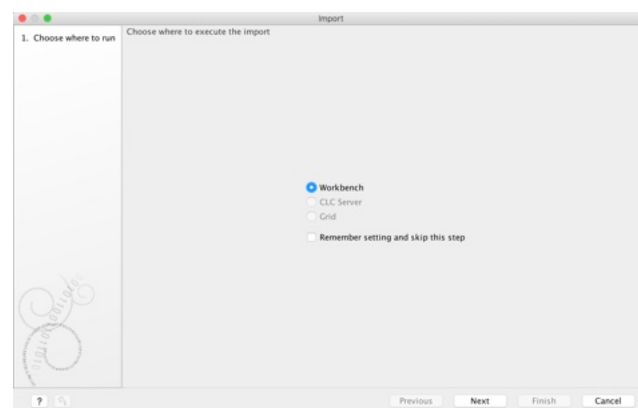
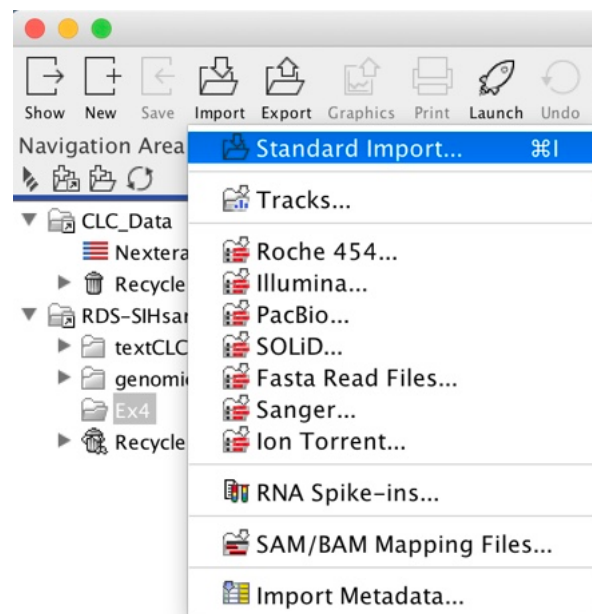
Explore genotype and phenotype

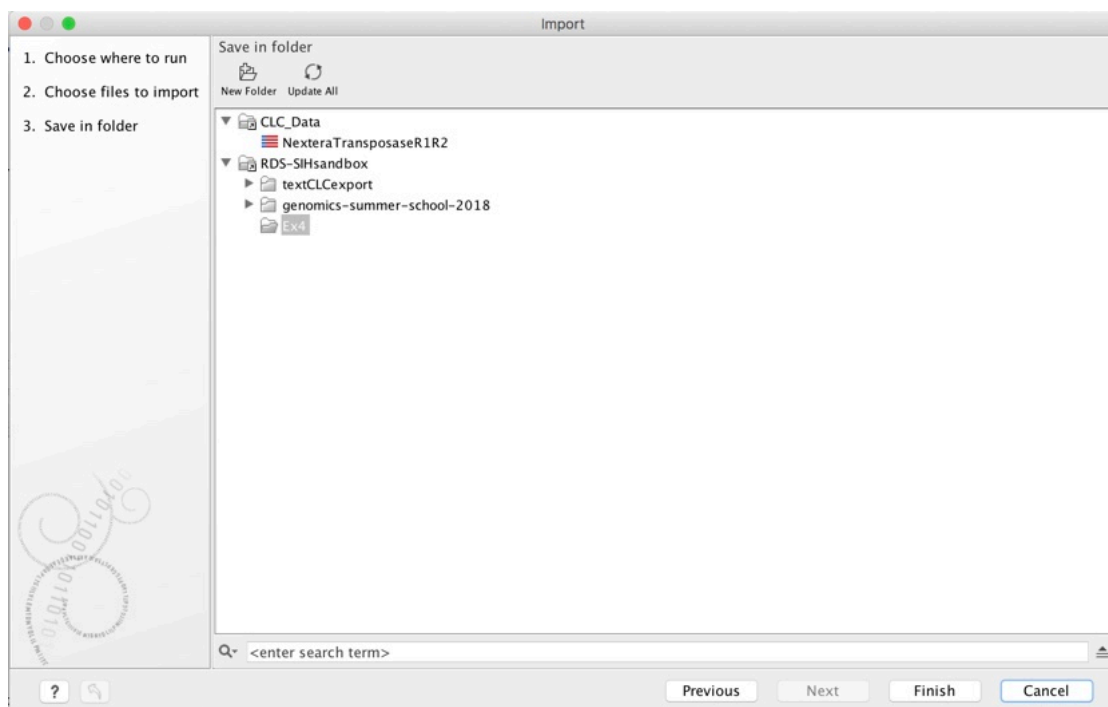
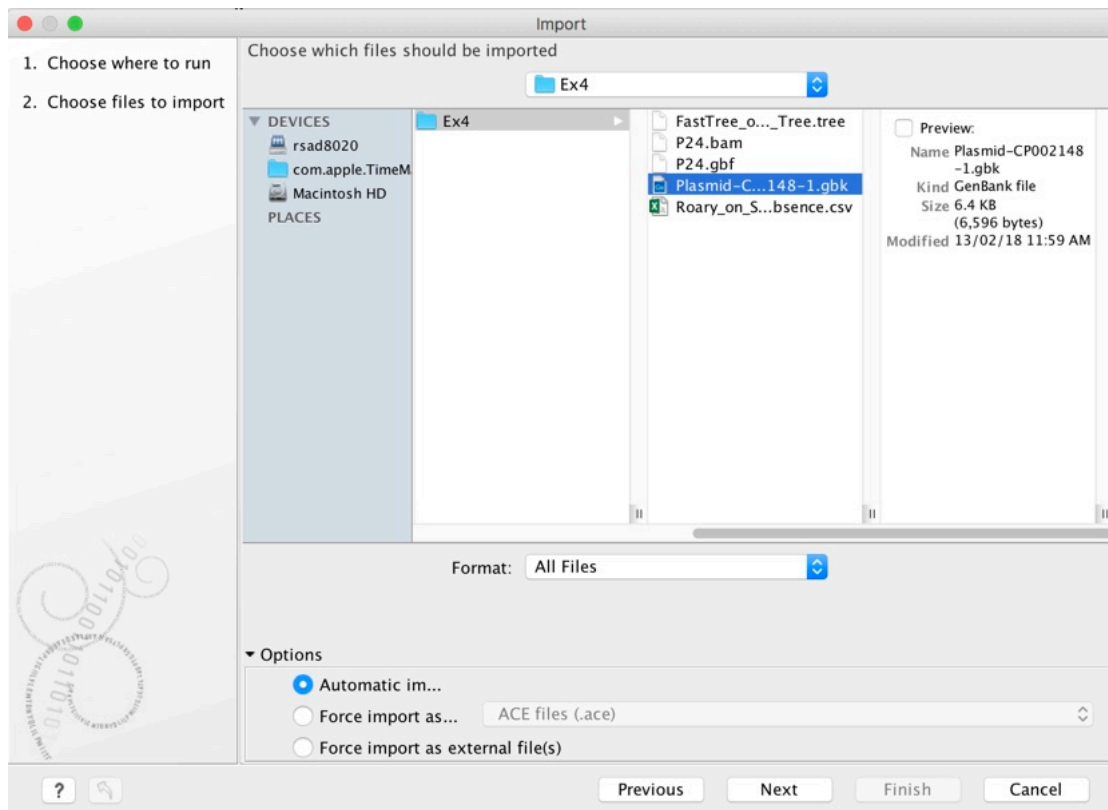
We will map our reads to the published plasmid carrying ermC sequence [CP002148](#)

1. Open CLC Genomics Workbench

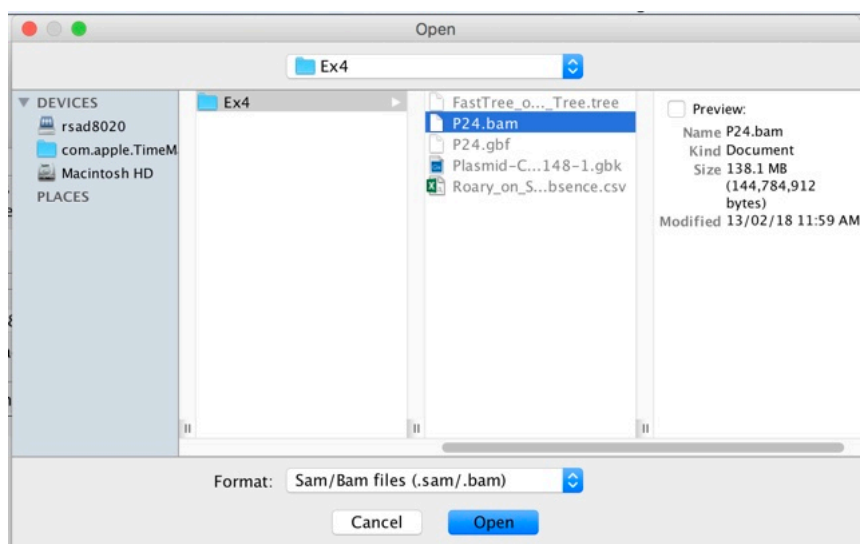
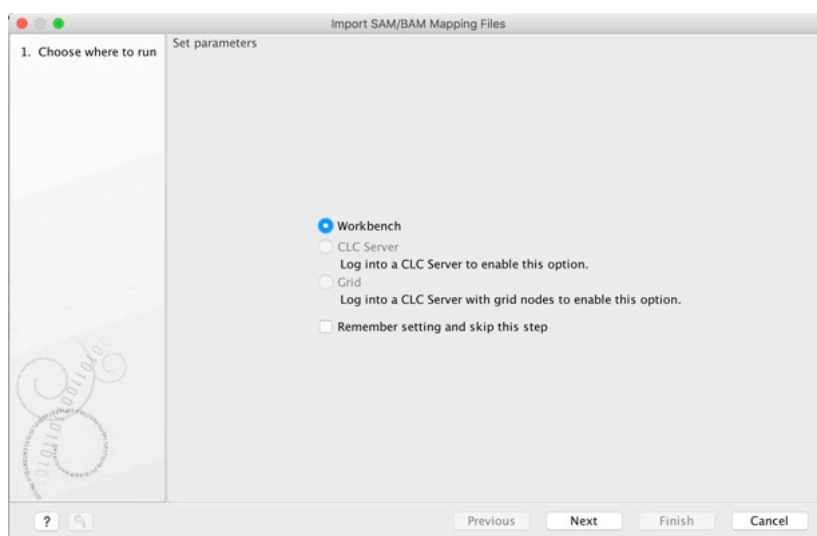
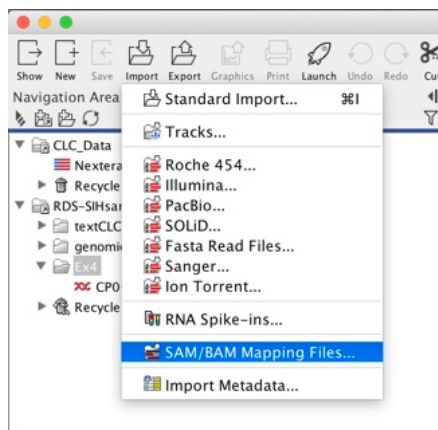


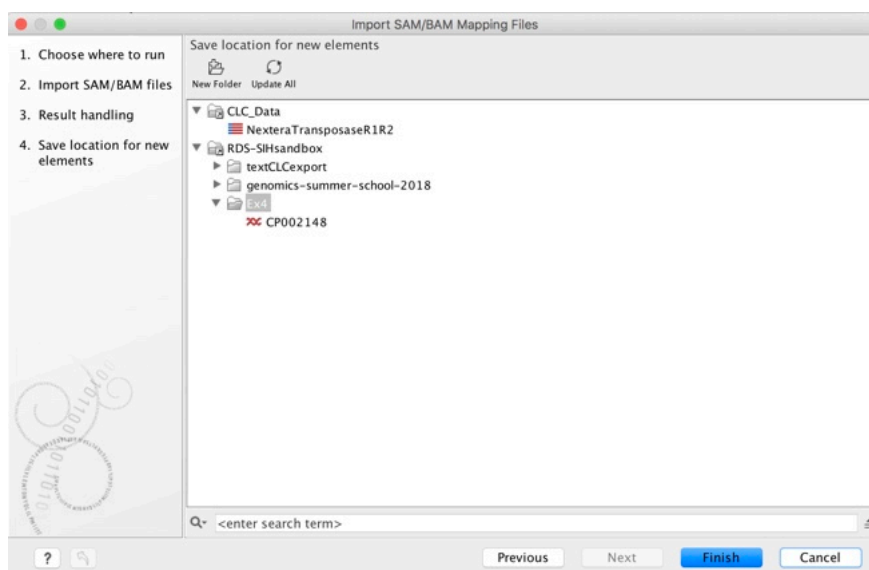
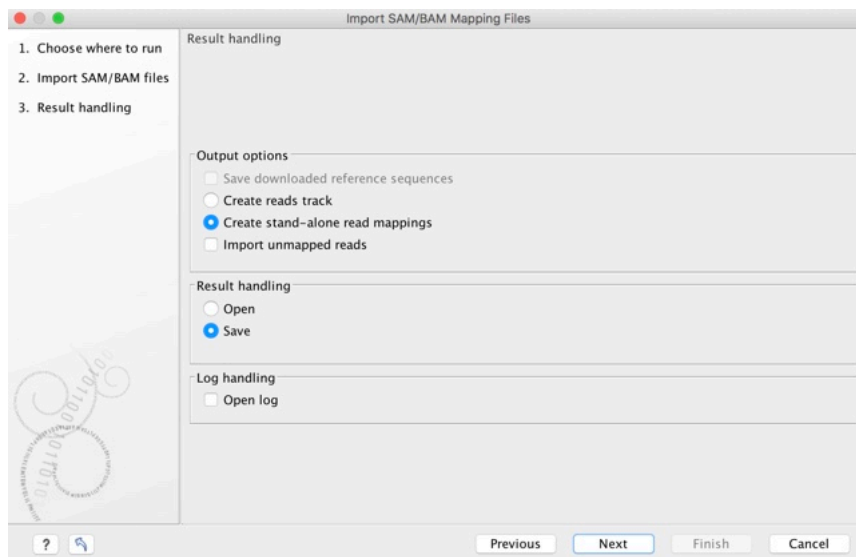
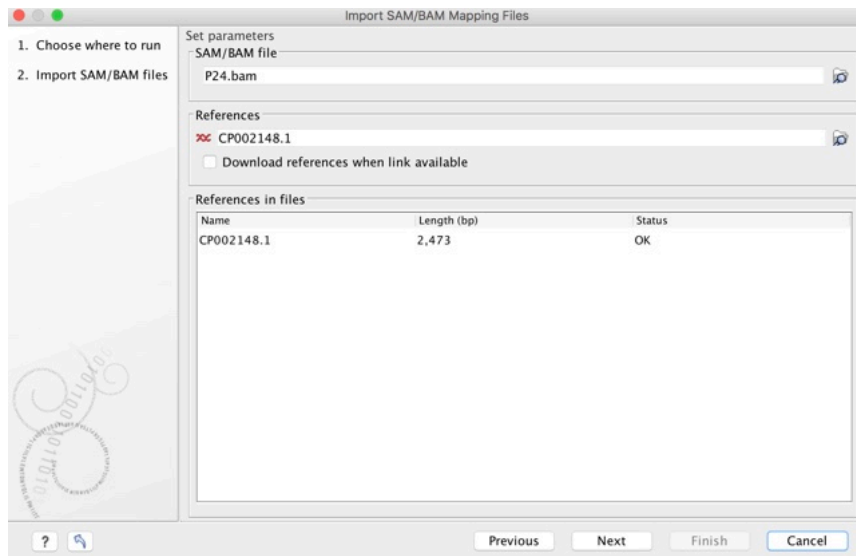
2. Import the reference plasmid carrying ermC sequence into CLC, [Plasmid-CP002148-1.gbk](#)

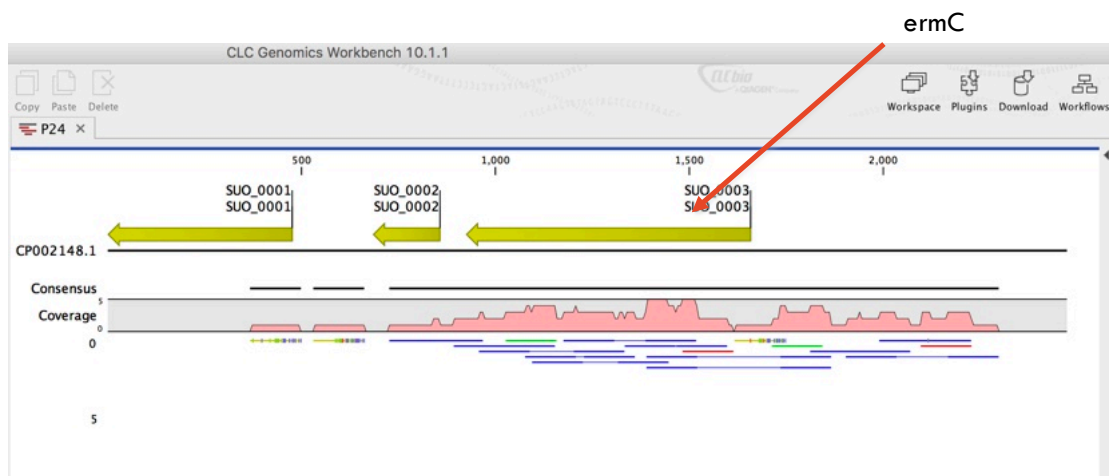




3. Import the mapping file **P24.bam** (note, this was generated for you)



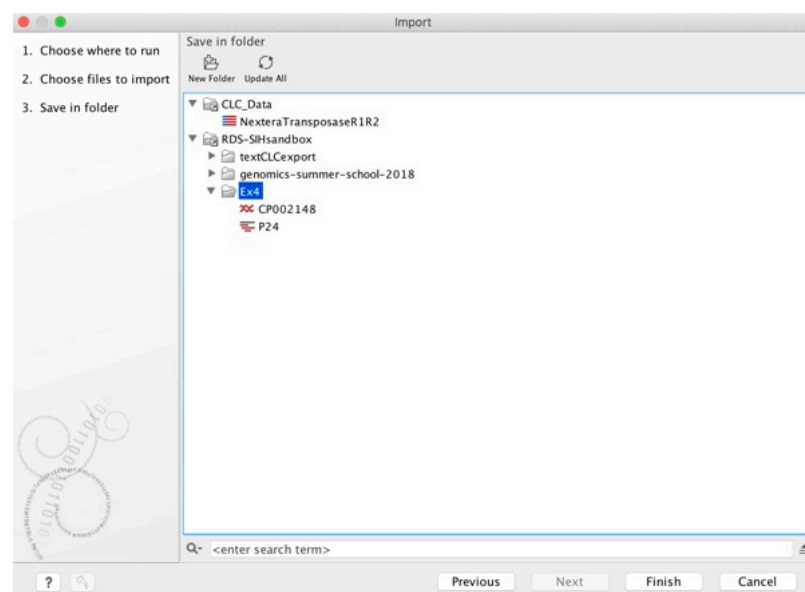
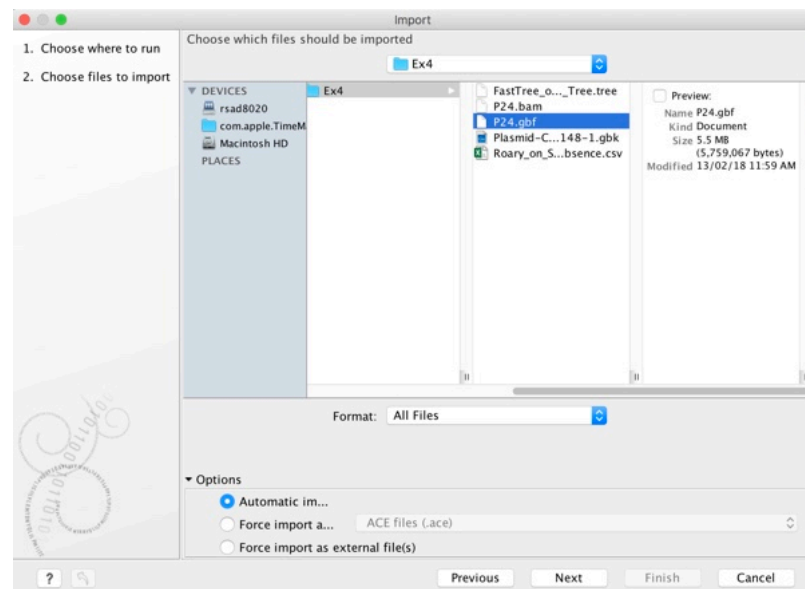
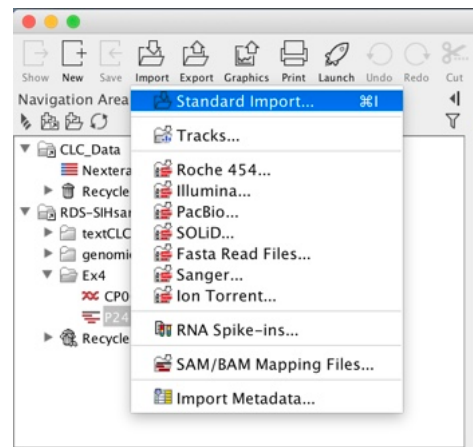




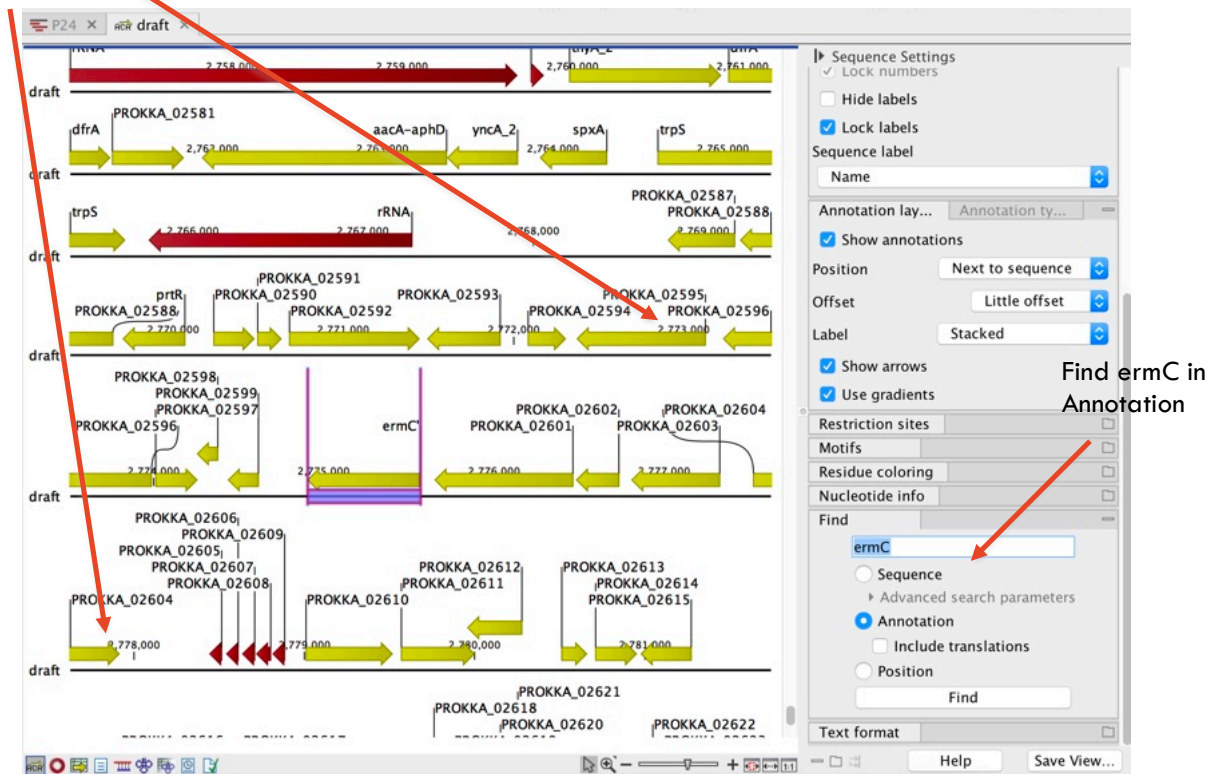
Notice that there is only partial coverage of the plasmid.

Let's look at the denovo assembly to see where the ermC gene that we detected is located.

1. Import the annotated draft genome (contigs stitched together) **P24.gb** into CLC.



Transposase/IS
protein



Discussion: Did the *ermC* gene move from the plasmid to the chromosome through a transposase?

Thank you



CIDM – PH Pathogen Genomics Team



Sydney Informatics Hub – Research Computing team
bioinformatics, modelling, simulation

The Sydney Informatics Hub is a Core Research Facility. We provide support, training, and advice on research data, analyses and computing. We also collaborate on grants and projects. Talk to us about your computing infrastructure, digital tools and data governance needs. Contact sih.info@sydney.edu.au for more information.

<https://informatics.sydney.edu.au>



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—
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centre for infectious diseases & microbiology
Public Health

Marie Bashir Institute
Tackling infections, locally and globally

